

Model-Free Adaptive Control for Contact-Rich Robotic Grasping under Dynamics Uncertainty

CS5478 Intelligent Robots: Algorithms and Systems – Li Jiaqi

Motivation and Research Question

Robotic grasping requires not only reaching a desired object pose, but also controlling physical interaction after contact. While analytical controllers can efficiently generate stable reaching and nominal grasping motions, successful grasping additionally depends on physical properties such as object mass, friction, contact forces, and possible slippage. These properties may be unknown or vary between objects, causing a controller designed for nominal dynamics to degrade under dynamics shifts.

Model-free reinforcement learning provides a complementary approach in which corrective control behavior can be learned through interaction with the environment. Rather than requiring a reinforcement learning policy to learn low-level stabilization and manipulation entirely from scratch, a learned residual policy can augment a nominal controller with torque corrections based on proprioceptive and contact feedback. This allows learning to focus on interaction-dependent adaptation while retaining the stability and efficiency of conventional control.

This project investigates whether a **model-free residual reinforcement learning policy can learn adaptive torque corrections for robotic grasping under uncertain object dynamics**.

Research Question: Can a model-free residual policy use interaction feedback to adapt its grasping actions to unseen object dynamics, such as variations in mass and friction, without explicit knowledge of these physical parameters?

Proposed Method

A 7-DoF robotic arm with a parallel gripper will be simulated in MuJoCo. A nominal controller will provide basic arm stabilization, reaching, and grasping behavior. The reinforcement learning policy will augment this controller by producing residual torque commands.

The policy will observe robot and task states including joint positions q , joint velocities $\dot{q}$, end-effector state, object pose and velocity, and finger contact/force measurements. Importantly, physical parameters such as object mass m and friction coefficient μ will **not** be provided directly to the policy.

At each control step, the learned policy outputs residual torque commands,

$$a_t = \Delta\tau_t = [\Delta\tau_1, \Delta\tau_2, \dots, \Delta\tau_7, \Delta\tau_{\text{grripper}}],$$

which are combined with the nominal controller output,

$$\tau_t = \tau_{\text{nominal}}(s_t) + \Delta\tau_t.$$

The residual actions will be bounded to prevent the learned policy from producing excessively large corrections. Soft Actor-Critic (SAC) will be used as the primary model-free continuous-control algorithm.

Training will follow a staged curriculum:

- **Stage 1 – Nominal Controller Validation:** Implement and validate stable reaching and basic grasping using the nominal controller. This stage establishes the underlying control pipeline and provides the reference behavior for later experiments.
- **Stage 2 – Residual Learning under Nominal Dynamics:** Train SAC to produce bounded residual torque corrections while performing grasp-and-lift tasks under fixed nominal object dynamics.
- **Stage 3 – Dynamics Randomization:** Train the residual policy with randomized object mass, friction, and initial object position to encourage robust corrective behavior under dynamics uncertainty.

Reward shaping will combine reaching distance, stable contact, object lift height, successful grasp completion, and a penalty on excessive residual actions. Penalizing large residual torques encourages the learned policy to rely on the nominal controller when no correction is necessary.

To evaluate generalization rather than only performance within the training distribution, the learned policy will also be tested on mass and friction conditions not encountered during training. Since these physical parameters are hidden from the policy, changes in residual control behavior must be driven by observable robot–object interaction feedback rather than direct access to the underlying dynamics parameters.

Baselines and Experimental Evaluation

The primary baseline will be the nominal controller without learned residual corrections. It will use conventional feedback control for arm motion together with contact-aware gripper control. This provides a direct reference for measuring the contribution of the learned residual policy.

The learned controller uses the same nominal control structure but augments it with SAC residual torques,

$$\tau_t = \tau_{\text{nominal}}(s_t) + \pi_{\theta}(s_t).$$

This design enables a controlled comparison between conventional feedback control and feedback control augmented by learned interaction-dependent corrections.

The core evaluation will compare the nominal controller and the residual SAC controller under both nominal dynamics and unseen variations in object mass and friction. Additional ablation experiments, if time permits, will examine the roles of contact feedback and dynamics randomization.

Experiment	Comparison	Metric	Purpose
Nominal Control	Nominal controller alone	Position error, grasp success rate	Validate baseline control pipeline
Residual Learning	Nominal controller vs. residual SAC	Grasp success rate, residual magnitude	Measure benefit of learned correction
Dynamics Shift	Nominal controller vs. residual SAC under unseen mass/friction	Grasp success rate, performance degradation	Evaluate adaptation to hidden dynamics
Contact Feedback	Residual SAC with vs. without contact observations	Grasp success rate	Measure value of interaction feedback
Dynamics Randomization	Fixed-dynamics vs. randomized-dynamics residual SAC	Unseen-condition success rate	Evaluate generalization

Expected Contribution

This project aims to provide an empirical study of model-free residual control for contact-rich robotic grasping under uncertain object dynamics. The main contribution is a systematic evaluation of whether a learned residual torque policy can use interaction feedback to improve grasping robustness when object mass and friction differ from nominal conditions, without direct access to these physical parameters.

By comparing a conventional nominal controller with the same controller augmented by SAC residual corrections, the project isolates the contribution of learned adaptive behavior while retaining a common underlying control structure. Evaluation on unseen mass and friction conditions will test whether the residual policy can generalize beyond the dynamics encountered during training.

Additional ablation experiments will investigate whether contact observations and dynamics randomization contribute to the robustness and generalization of the learned policy. Analysis of the residual actions may further reveal whether the policy applies larger or systematically different corrections when interaction conditions depart from nominal dynamics.

Scope and Risk Management

The minimum complete project will include implementation of a stable nominal reaching-and-grasping controller, residual SAC training under nominal dynamics, evaluation under unseen variations in object mass and friction, and comparison between the nominal and residual controllers.

The contact-feedback and dynamics-randomization ablations will be treated as additional experiments if time permits. These experiments will investigate whether robustness under dynamics uncertainty can be attributed specifically to interaction feedback and randomized training rather than nominal task learning alone.

A recurrent residual policy may be explored as a further extension to determine whether interaction history helps the agent infer hidden physical properties from previous contacts and improve subsequent corrective actions.

Using residual rather than unrestricted raw torque control reduces the difficulty of learning basic 7-DoF stabilization from scratch and allows the reinforcement learning component to focus on interaction-dependent corrections. The project will therefore progress from nominal controller validation to residual learning and finally to uncertain dynamics. This staged structure ensures that each stage provides an independently evaluable result while progressively addressing the main research question.